Evaluation: checks/mnist-smoke/validation-check

Purpose and interpretation

This evaluation measures a fixed checkpoint and fixed noise population. KID subset SD describes metric subsampling variability; it is not a confidence interval across trained seeds. The first 64 images are shown without selection.

Recorded evidence


```
{
  "dataset": "mnist",
  "family": "onn",
  "training_seed": 101,
  "split": "validation",
  "intervention": "none",
  "generated_count": 128,
  "real_count": 5000,
  "sampling_seed": 20261016,
  "metric_seed": 20261016,
  "nfe": 32,
  "kid_subsets": 100,
  "kid_subset_size": 128,
  "checkpoint_sha256":
"8f38d4c79f09a651583cc67b6596ab438a75269ad8fcb6a89db4135257a2a9db",
  "data_sha256":
"39da09b6ccbbfae8c6424de59b611882eba7f0fb590999ef1851e80f2d027b1c",
  "source_revision": "5b69c903ddac223b5d44e6967dcd49ebd4e136cc",
  "primary_eligible": false,
  "device": "cuda",
  "status": "completed",
  "sampling": {
    "count": 128,
    "seed": 20261016,
    "nfe": 32,
    "batch_size": 32,
    "seconds": 2.681424114998663,
    "seconds_per_image": 0.020948625898427053,
    "precision": "float32",
    "timing_excludes": "model load and warmup",
    "clipped_fraction": 0.3916613459587097,
    "peak_memory_bytes": 55247872
  },
  "metrics": {
    "pixel_sliced_wasserstein": 1.1010091948096694,
    "projections": 128
  },
  "validation_velocity_mse": 2.0086912736296654,
  "metric_seconds": 0.24426259099709569,
  "id": "checks/mnist-smoke/validation-check",
  "artifact": "runs/checks/mnist-smoke/validation-check/evaluation.json"
}
```

Fixed sample grid

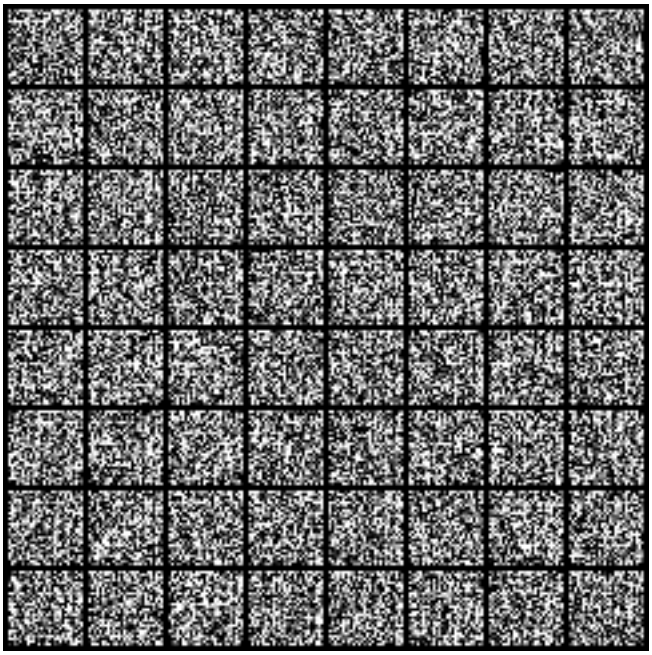


Figure 1: First 64 generated samples

Method choices and learning

The shared optimizer, data policy, architecture, and stopping rule were chosen before final comparison and are described in the technical paper. The 3e-4 learning rate is a conservative AdamW starting point, not a tuned optimum. Microbatches bound laptop memory; four internal updates bound recurrent cost; EMA smooths noisy parameter updates. These choices trade study breadth for a fixed local compute budget. This record supports only the measurements above; causes of quality differences require the trained controls and uncertainty analysis. Full trajectories, populations, RNG states, and logs remain in the referenced artifact bundle. Sources: protocol, TorchCFM, AKOrN.